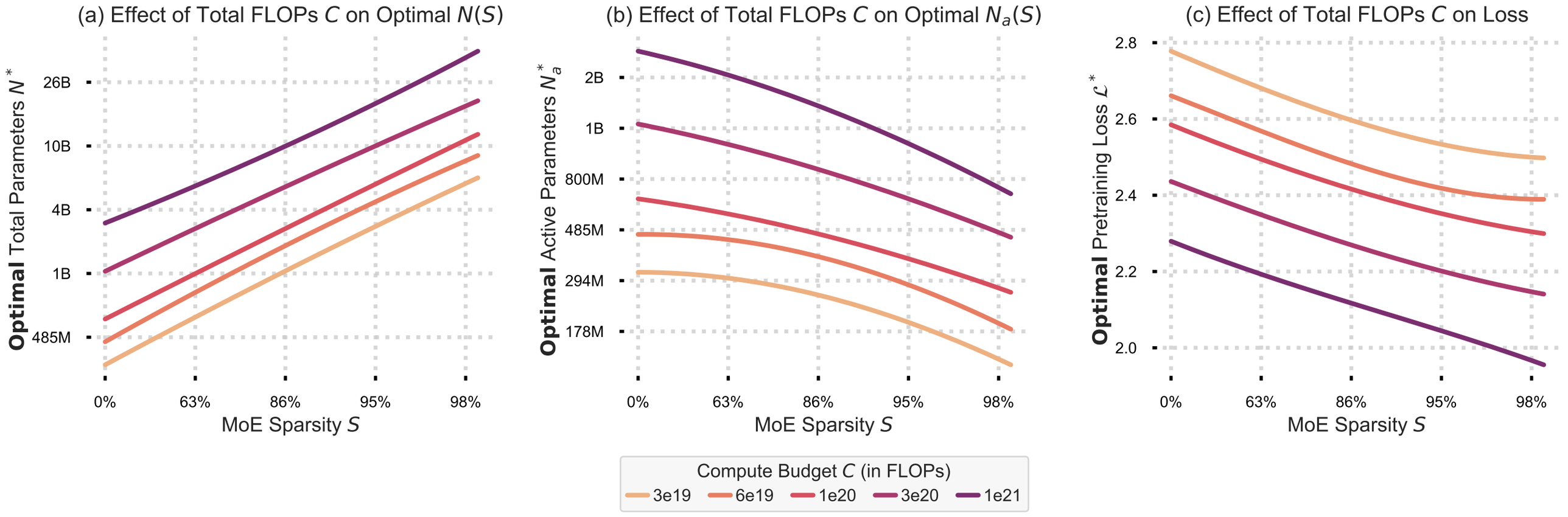
(a) Effect of Total FLOPs C on Optimal N(S)
Optimal Total Parameters N*
26B
10B
4B
1B
485M
0%
63%
86%
95%
98%
MoE Sparsity S
(b) Effect of Total FLOPs C on Optimal N_a(S)
Optimal Active Parameters N_a*
2B
1B
800M
485M
294M
178M
0%
63%
86%
95%
98%
MoE Sparsity S
(c) Effect of Total FLOPs C on Loss
Optimal Pretraining Loss L*
2.8
2.6
2.4
2.2
2.0
0%
63%
86%
95%
98%
MoE Sparsity S
Compute Budget C (in FLOPs)
3e19
6e19
1e20
3e20
1e21